MIRPUE UNIVERSITY OF SCIENCE & TECHNOLOGY MUST MIRPURE (AJ&K)

Assignment 1

COURSE TITLE:	Cloud Computing
SEMESTER:	4th Semester (23-27)
SECTION:	(Section B)
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1. AVR C Source Code:

```
#include <avr/io.h>

int main(void)
{
   /* Pin Direction Configuration */
   DDRA = 0x80;    /* PA7 as output → Motor */
   DDRB = 0x01;    /* PB0 as output → Status LED */
   DDRC = 0x00;    /* PC5 as input  → Door switch */

   while (1)
   {
      /* Read Door Status */
      if (PINC & 0x20)       /* Door OPEN */
      {
         PORTA = 0x80;      /* Motor ON */
         PORTB = 0x01;      /* LED ON */
      }
      else                  /* Door CLOSED */
      {
         PORTA = 0x00;      /* Motor OFF */
         PORTB = 0x00;      /* LED OFF */
      }
   }
}
```

2. Logic Explanation:

Introduction:

The Intelligent Automatic Door System is an embedded control application designed using an AVR microcontroller. Its purpose is to automatically close a door when it is detected as open and to provide a visual indication of the door status for safety. The system operates using a simple and efficient control strategy to ensure reliable performance without unnecessary motor operation.

The design strictly uses low-level AVR input/output registers (`DDRx`, `PORTx`, and `PINx`) to meet embedded system constraints and to achieve predictable, real-time behavior.

Purpose of the System:

The main purpose of this system is to implement intelligent door control by continuously monitoring the door condition and responding appropriately. The system ensures that:

- The door is automatically closed if left open.
- The motor operates only when required.
- A visual indicator reflects the current door status.
- Repeated or unnecessary motor activation is avoided.

Hardware Interface Description:

The system interfaces with three external components connected to specific AVR pins:

Component	AVR Pin	Function
Door Motor (through driver)	PA7	Controls door closing
Door Status Switch	PC5	Detects door position
Status LED	PB0	Indicates door status

The door status switch provides a digital signal where:

- Logic `1` indicates that the door is open.
- Logic `0` indicates that the door is closed.

I/O Register Configuration:

- **Data Direction Setup**

  - **DDRA = 0x80** configures PA7 as an output pin to control the motor.
  - **DDRB = 0x01** configures PB0 as an output pin for the status LED.
  - **DDRC = 0x00** configures Port C as input, allowing PC5 to read the door state.

Continuous Door Monitoring:

The system operates within an infinite loop, enabling continuous monitoring of the door status. This ensures that any change in the door position is detected immediately.

The door switch state is read using the `PINC` register. A bitwise AND operation with `0x20` is used to check the condition of pin PC5 without affecting other bits.

Control Logic Operation:

- **Door Open Condition**

When the door switch input reads logic `1`:

- The motor control pin (PA7) is set high, turning the motor ON to close the door.
- The status LED (PB0) is turned ON to indicate that the door is open.

Door Closed Condition

When the door switch input reads logic `0`:

- The motor control pin (PA7) is cleared, turning the motor OFF.
- The status LED (PB0) is turned OFF.

This ensures the motor stops as soon as the door reaches the closed position.

Intelligent Purpose-Based Control:

The system follows a purpose-based control strategy:

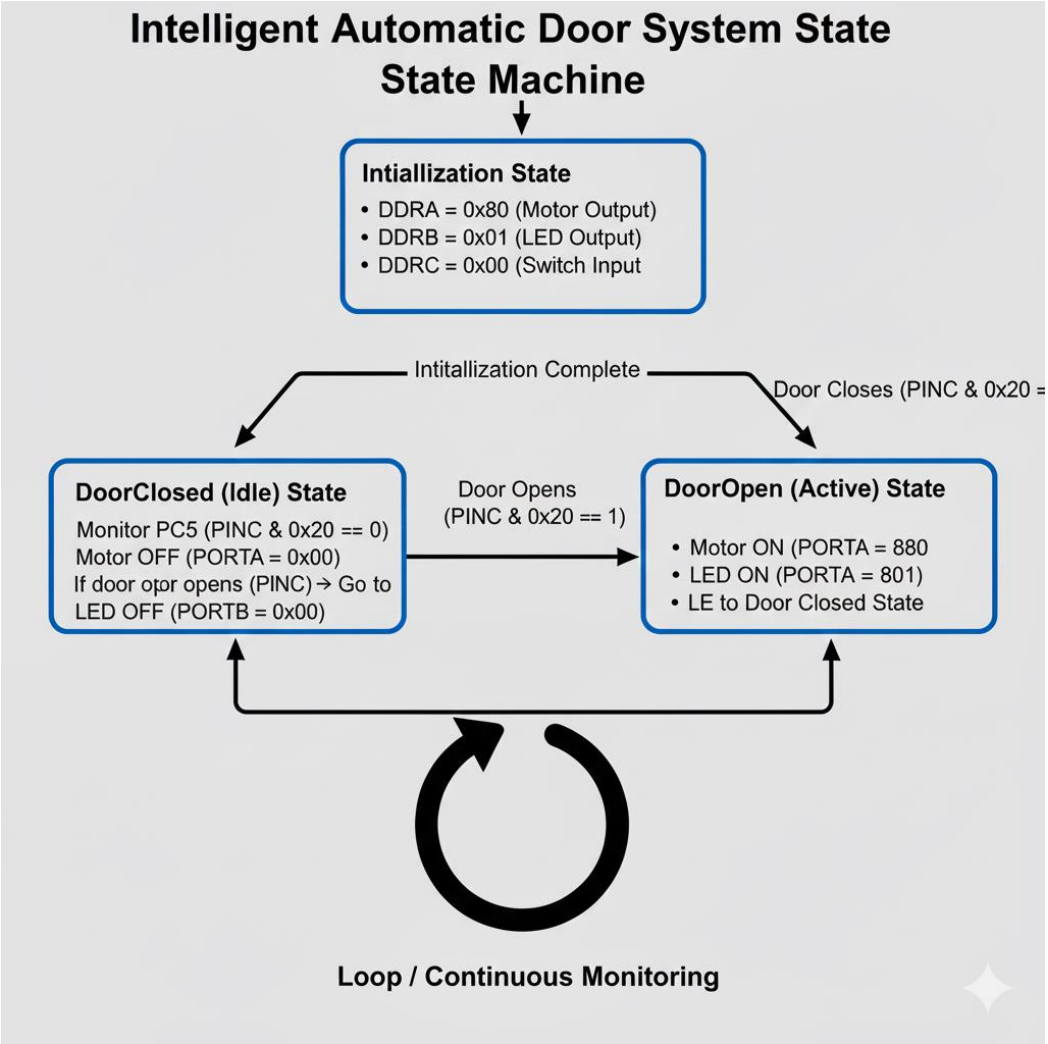
- The motor is activated only when the door is open.
- Once the door is closed, the motor remains off.
- The motor is not reactivated unless the door opens again.

This behavior prevents unnecessary actuation and enhances system efficiency.

Safety and Reliability Features:

- Continuous monitoring improves responsiveness.
- Simple logic reduces software complexity and failure risk.
- Direct register-level control ensures fast and predictable operation.
- The LED provides clear visual feedback for user safety

3.Flowchart or State Diagram:



4.Group Contribution Table: